

Size-Aware In-Pixel Transimpedance Amplifier Design

Bioelectronics Assignment (Resit)

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1 Core Transimpedance Amplifier without Gain Boosting

This section presents the design, operating point, and small-signal performance of the core in-pixel transimpedance amplifier (TIA) without the gain-boosting amplifier. The objective of this section is to verify the basic functionality of the current-conveyor-based TIA and to identify its limitations prior to introducing gain boosting.

1.1 Circuit Architecture

Figure 1 shows the schematic of the core TIA. The circuit consists of an NMOS current conveyor formed by transistors M1 and M2, and a diode-connected PMOS load M3 that converts the input current into an output voltage. The electrode-electrolyte interface is modeled by a Norton equivalent consisting of a current source in parallel with a resistance of 10 M Ω and a capacitance of 0.5 pF.

The output node V_{TIA} is loaded with a capacitance $C_L = 1$ pF, representing the pixel output load. The supply voltage is $V_{DD} = 1.8$ V.

1.2 Device Sizing and Bias Conditions

The transistor dimensions used in this confirming design stage are listed below:

- M1 (NMOS): $W = 2 \mu\text{m}$, $L = 0.18 \mu\text{m}$
- M2 (NMOS): $W = 1 \mu\text{m}$, $L = 0.18 \mu\text{m}$
- M3 (PMOS): $W = 2 \mu\text{m}$, $L = 0.5 \mu\text{m}$

The gate bias voltages are set to $V_{B1} = 0.5$ V for M1 and $V_{M2,\text{bias}} = 0.7$ V for M2, resulting in a nominal bias current of approximately 5 μA through the TIA core.

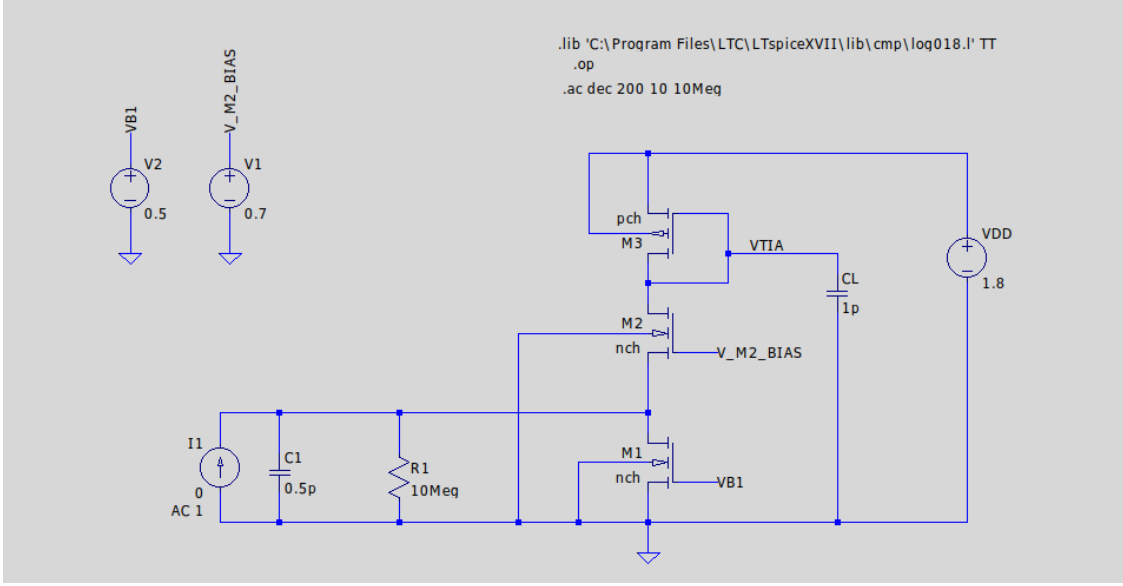


Figure 1: Core TIA schematic without gain boosting.

1.3 DC Operating Point

The DC operating point was extracted using a `.op` simulation. The resulting node voltages and currents are summarized below:

- $V_{TIA} = 1.147$ V
- $V_{in} = V(n002) = 0.144$ V
- $I_{DD} = |I(V_{DD})| = 4.95$ μ A
- $I_D(M1) \approx I_D(M2) \approx I_D(M3) \approx 4.95$ μ A

The DC input voltage remains small, confirming correct current-conveyor operation, while sufficient output headroom is preserved at V_{TIA} .

1.4 Gain-Boosting Amplifier Transconductance Efficiency (g_m/I_D)

The transconductance efficiency of the gain-boosting devices (M4–M8) was extracted from the operating-point report in Fig. 9. Table 1 lists the measured drain currents, transconductances, and g_m/I_D values.

1.5 Small-Signal Transimpedance Gain

The transimpedance gain was obtained from an AC simulation by exciting the input with a 1 A AC current source and plotting V_{TIA} . The low-frequency transimpedance gain is measured as:

--- Operating Point ---			
V(n001):	1.8	voltage	
V(vtia):	1.14741	voltage	
V(v_m2_bias):	0.7	voltage	
V(n002):	0.143898	voltage	
V(vb1):	0.5	voltage	
Id(M1):	4.93407e-006	device_current	
Ig(M1):	0	device_current	
Ib(M1):	-1.53859e-013	device_current	
Is(M1):	-4.93407e-006	device_current	
Id(M2):	4.94847e-006	device_current	
Ig(M2):	0	device_current	
Ib(M2):	-9.98363e-012	device_current	
Is(M2):	-4.94846e-006	device_current	
Id(M3):	-4.94847e-006	device_current	
Ig(M3):	-0	device_current	
Ib(M3):	6.62595e-013	device_current	
Is(M3):	4.94847e-006	device_current	
I(C1):	7.19488e-026	device_current	
I(C1):	1.14741e-024	device_current	
I(I1):	0	device_current	
I(R1):	1.43898e-008	device_current	
I(V2):	0	device_current	
I(V1):	0	device_current	
I(Vdd):	-4.94847e-006	device_current	

Figure 2: DC operating point results of the core TIA.

Table 1: g_m/I_D extraction for gain-boosting amplifier devices (from Fig. 9).

Device	I_D [μ A]	g_m [μ S]	g_m/I_D [V^{-1}]
M4	1.30	79.1	60.85
M5	0.658	14.6	22.19
M6	0.645	14.3	22.17
M7	0.658	11.4	17.33
M8	0.645	11.2	17.36

$$G_{TIA,LF} = 86.78 \text{ dB}\Omega \approx 21.8 \text{ k}\Omega$$

This value satisfies the specification requirement of $G_{TIA} > 10 \text{ k}\Omega$. The dominant contribution to the transimpedance gain is the diode-connected PMOS load M3, such that:

$$G_{TIA} \approx \frac{1}{g_{m3}}$$

The -3 dB bandwidth is observed at approximately:

$$\text{BW} \approx 6.4 \text{ MHz}$$

which comfortably exceeds the minimum required bandwidth of 100 kHz .

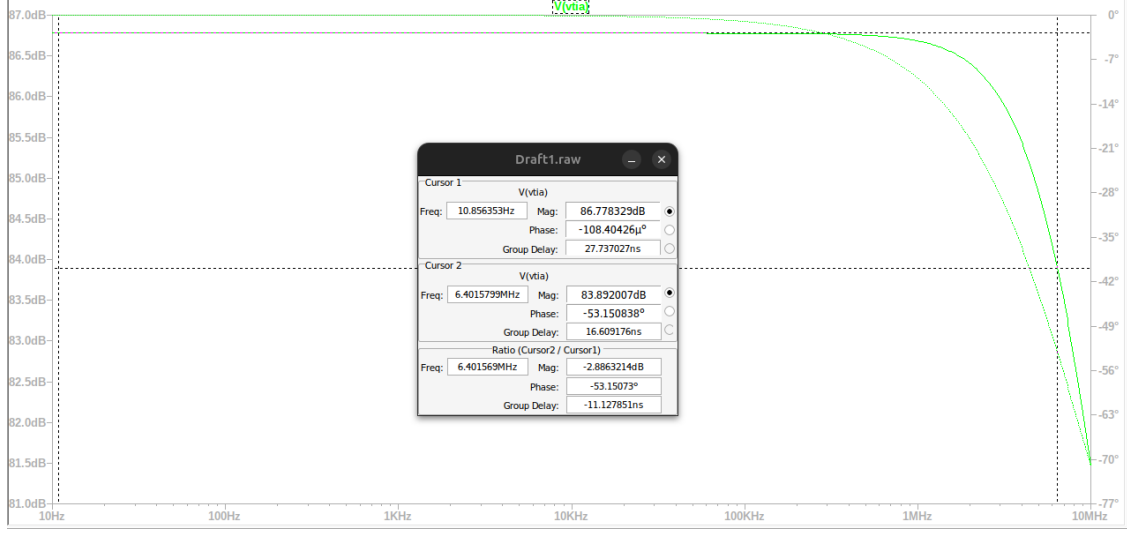


Figure 3: AC transimpedance gain magnitude of the core TIA.

1.6 Input Impedance

The input impedance was evaluated by plotting the input node voltage $V(n002)$ under a 1 A AC current excitation. The low-frequency input impedance is measured as:

$$Z_{in,LF} = 79.12 \text{ dB}\Omega \approx 9.0 \text{ k}\Omega$$

This value does not meet the specification requirement of $Z_{in} < 1 \text{ k}\Omega$. The relatively high input impedance is expected in the absence of gain boosting, as it is primarily set by the finite transconductance of the input transistor M1.



Figure 4: Input impedance magnitude of the core TIA.

1.7 Transconductance Efficiency

The transconductance and transconductance efficiency of the core transistors were extracted from the operating point simulation and are summarized in Table 2.

Table 2: Transistor operating point and transconductance efficiency.

Device	I_D [μA]	g_m [μS]	g_m/I_D [V^{-1}]
M1	4.93	92.8	18.8
M2	4.95	84.6	17.1
M3	4.95	42.9	8.67

The lower g_m/I_D value of M3 is consistent with its role as a load device optimized for transimpedance rather than input impedance.

1.8 Discussion

While the core TIA meets the transimpedance gain and bandwidth specifications, the input impedance remains significantly above the required limit. This motivates the introduction of a gain-boosting amplifier to increase the effective loop gain and reduce the input impedance in the next design stage.

2 Transimpedance Amplifier with Gain Boosting

This section extends the core transimpedance amplifier presented in Section 1 by introducing a gain-boosting amplifier around the NMOS current conveyor. The purpose of the gain-boosting stage is to increase the effective loop gain of the input transistor and thereby reduce the input impedance while preserving transimpedance gain, bandwidth, and noise performance.

2.1 Circuit Architecture with Gain Boosting

Figure 5 shows the complete TIA schematic including the gain-boosting amplifier. The core TIA formed by transistors M1–M3 is retained, while an auxiliary amplifier consisting of transistors M4–M8 senses the cascode node voltage V_{CAS} and drives the gate of M2. This configuration increases the small-signal output resistance seen at the drain of M1, thereby enhancing the effective loop gain of the current conveyor.

The electrode–electrolyte interface remains modeled by a Norton equivalent consisting of a current source in parallel with $R = 10 \text{ M}\Omega$ and $C = 0.5 \text{ pF}$. The output load capacitance is kept at $C_L = 1 \text{ pF}$, and the supply voltage remains $V_{DD} = 1.8 \text{ V}$.

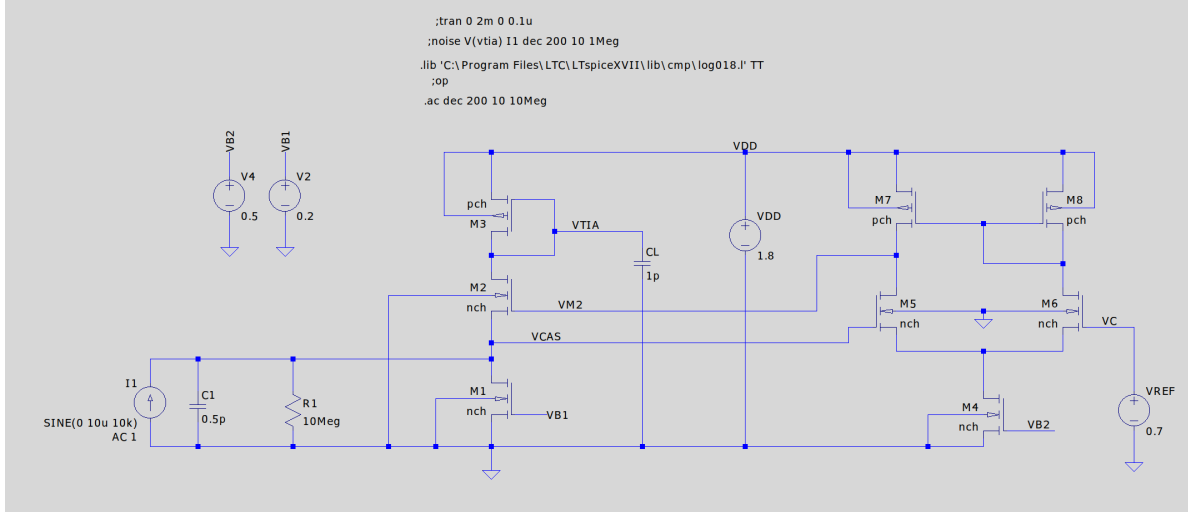


Figure 5: Complete TIA schematic including the gain-boosting amplifier.

2.2 Explicit Transfer Function V_{TIA}/I_{IN} (with and without gain boosting)

Case 1: Core TIA without gain boosting. At node V_{TIA} the dominant small-signal element is the diode-connected PMOS load (M3), whose incremental resistance is

$$r_{d3} = \frac{1}{g_{m3} + g_{ds3}} \approx \frac{1}{g_{m3}}.$$

With the output load capacitor C_L , the transimpedance transfer is well approximated by a single-pole form:

$$\frac{V_{TIA}(s)}{I_{IN}(s)} \approx \frac{r_{d3}}{1 + s r_{d3} C_L}.$$

This matches the measured low-frequency transimpedance in Section 1, where

$$\left| \frac{V_{TIA}}{I_{IN}} \right|_{LF} \approx 21.8 \text{ k}\Omega,$$

and the dominant pole frequency is

$$\omega_{p,1} \approx \frac{1}{r_{d3} C_L}.$$

Case 2: TIA with gain boosting. With gain boosting enabled, the amplifier increases the effective loop gain around the current conveyor (mainly improving Z_{in}). To first order, the output transimpedance is still set by the small-signal resistance at V_{TIA} (again dominated by M3), thus

$$\frac{V_{TIA}(s)}{I_{IN}(s)} \approx \frac{r_{d3,GB}}{1 + s r_{d3,GB} C_L},$$

where

$$r_{d3,GB} = \frac{1}{g_{m3} + g_{ds3}}$$

evaluated at the boosted operating point and chosen M3 sizing.

From the AC result in Fig. 10, the low-frequency transimpedance is

$$\left| \frac{V_{TIA}}{I_{IN}} \right|_{LF} \approx 116.2 \text{ dB}\Omega \approx 64 \text{ k}\Omega,$$

and the measured -3 dB bandwidth is approximately

$$f_{-3dB} \approx 242 \text{ kHz}.$$

2.3 Gain-Boosting Amplifier Biasing and Open-Loop Configuration

To evaluate the intrinsic gain of the gain-boosting amplifier, the feedback path to M2 is opened and the amplifier is biased independently, as shown in Figure 6. A small-signal differential excitation of ± 0.5 V is applied to the amplifier input while maintaining identical DC bias conditions as in the closed-loop configuration.

2.4 Open-Loop Gain of the Gain-Boosting Amplifier

The open-loop gain A_{OL} of the gain-boosting amplifier was extracted from an AC simulation by plotting the ratio between the output control voltage V_C and the differential input voltage. The resulting magnitude and phase response are shown in Figure 7.

The low-frequency open-loop gain is measured as:

$$A_{OL,LF} \approx 34.4 \text{ dB} \approx 52 \text{ V/V}$$

with a dominant pole located above 100

2.5 Device Sizing and Area Considerations

The transistor dimensions used for the gain-boosted TIA are summarized in Table 3. The total active device area of the complete TIA, including the gain-boosting amplifier, is approximately $21.5 \mu\text{m}^2$, which comfortably satisfies the pixel area constraint of $50 \mu\text{m} \times 50 \mu\text{m}$.

2.6 DC Operating Point

The DC operating point of the gain-boosted TIA was verified using a .op simulation. The resulting node voltages are shown in Figure 8, with detailed device currents and operating regions reported in Figure 9.

Key DC results are summarized as follows:

- $V_{TIA} \approx 1.32 \text{ V}$

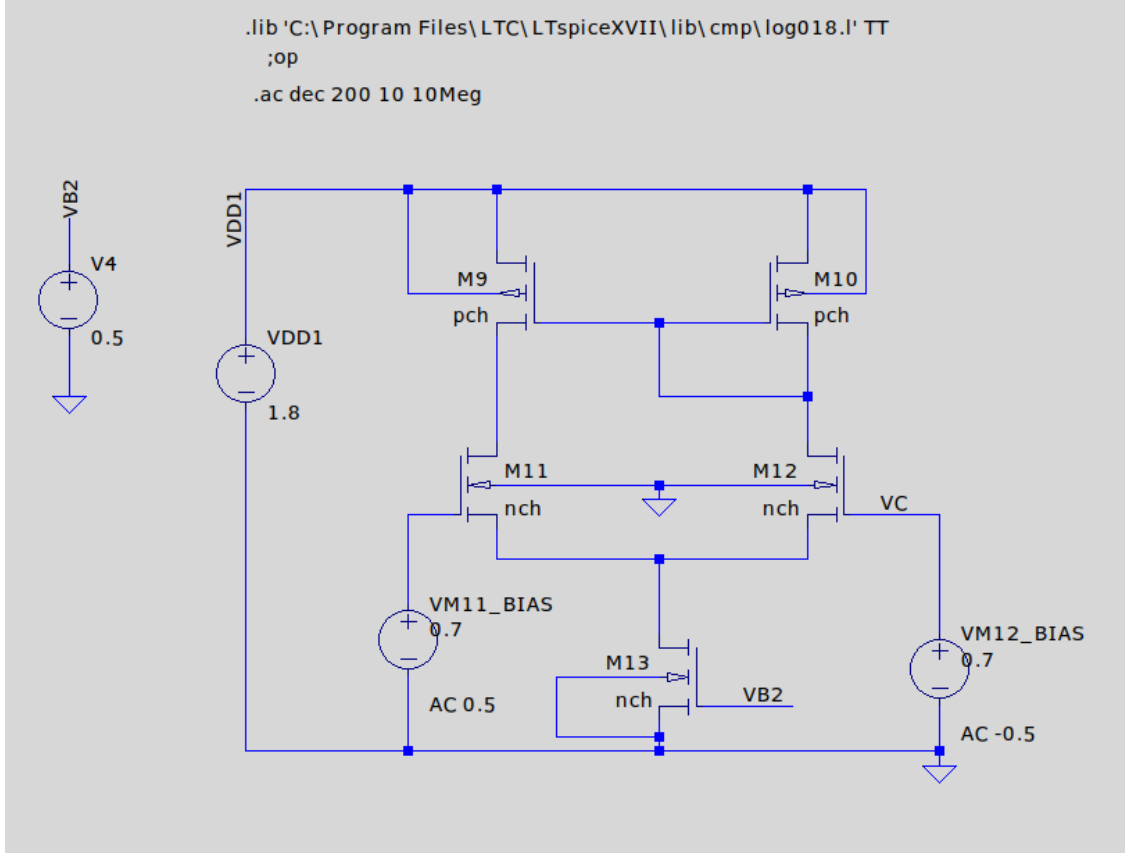


Figure 6: Biasing configuration used for open-loop gain extraction of the gain-boosting amplifier.

- $V_{in} = V(n002) \approx 0.22 \text{ V}$
- Total supply current $I_{DD} \approx 14 \mu\text{A}$

All transistors operate in saturation, confirming correct biasing of both the core TIA and the gain-boosting amplifier.

2.7 Transimpedance Gain and Bandwidth

The transimpedance gain was measured using an AC simulation with a 1 A input current excitation. Figure 10 shows the magnitude response of V_{TIA} .

The low-frequency transimpedance gain is:

$$G_{TIA,LF} \approx 116.2 \text{ dB}\Omega \approx 64 \text{ k}\Omega$$

with a measured -3 dB bandwidth of approximately 240 kHz, which remains well above the required minimum of 100 kHz.

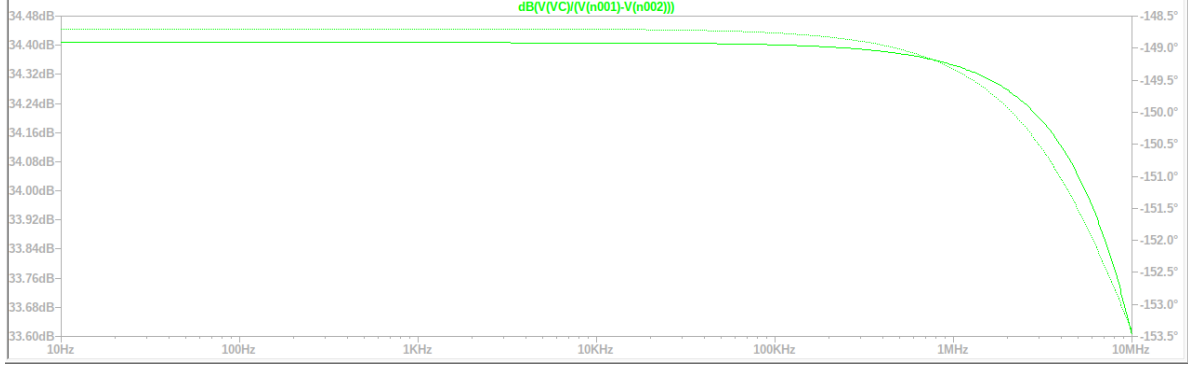


Figure 7: Open-loop gain magnitude and phase of the gain-boosting amplifier.

Table 3: Transistor sizing table for the gain-boosted TIA.

Device	Type	W (μm)	L (μm)	Active area $A = W \cdot L$ (μm^2)
M1	nch	20	0.5	10
M2	nch	10	0.5	5
M3	pch	1	2.0	2
M4	nch	1	0.5	0.5
M5	nch	2	0.5	1
M6	nch	2	0.5	1
M7	pch	2	0.5	1
M8	pch	2	0.5	1
Total				21.5

2.8 Input Impedance Reduction

The input impedance was evaluated by plotting the input node voltage $V(n002)$ under a 1 A AC excitation. Figure 11 shows the resulting magnitude response.

The low-frequency input impedance is measured as:

$$Z_{in,LF} \approx 58.1 \text{ dB}\Omega \approx 800 \Omega$$

which satisfies the specification requirement of $Z_{in} < 1 \text{ k}\Omega$. Compared to the unboosted case ($\sim 9 \text{ k}\Omega$), the gain-boosting amplifier achieves more than an order-of-magnitude reduction in input impedance.

2.9 Input-Referred Noise

The input-referred current noise was evaluated using a noise simulation with the input current source as the reference. Figure 12 shows the resulting spectral density.

Integrated over a bandwidth from 10 Hz to 1 MHz, the total RMS input-referred noise is:

$$I_{in,rms} \approx 814 \text{ pA}$$

which satisfies the specification requirement of $I_{in,rms} < 1 \text{ nA}$.

2.10 Transient Response

Figure 13 shows the transient output voltage V_{TIA} for a sinusoidal input current of $I_{in} = 10 \mu\text{A}$ at 10 kHz. The waveform exhibits stable operation with no evidence of ringing or distortion, confirming adequate phase margin of the gain-boosted loop.

2.11 Maximum Input Current $I_{in,\max}$

The maximum input current is limited by the available output swing at V_{TIA} before the circuit leaves its linear operating region (i.e., before headroom constraints cause noticeable gain compression or device triode operation). From the operating point in Fig. 8, the nominal output DC level is

$$V_{TIA,DC} \approx 1.319 \text{ V}.$$

From the transient response in Fig. 13 with a $10 \mu\text{A}$ sinusoidal excitation, the lowest observed output voltage is approximately

$$V_{TIA,\min} \approx 0.40 \text{ V},$$

suggesting an available downward swing of

$$\Delta V_{\downarrow} \approx V_{TIA,DC} - V_{TIA,\min} \approx 0.919 \text{ V}.$$

Using the measured low-frequency transimpedance gain (Fig. 10),

$$G_{TIA} \approx 64 \text{ k}\Omega,$$

the corresponding maximum input current magnitude before the output reaches this swing limit is

$$I_{in,\max} \approx \frac{\Delta V_{\downarrow}}{G_{TIA}} \approx \frac{0.919}{64 \cdot 10^3} \approx 14.4 \mu\text{A}.$$

To maintain margin against nonlinearity and process variation, a conservative value of

$$I_{in,\max} \approx 10 \mu\text{A}$$

is adopted, which is also directly validated by the stable transient waveform in Fig. 13.

2.12 Discussion

The introduction of the gain-boosting amplifier significantly improves the input impedance of the TIA while preserving transimpedance gain and noise performance. Although the total current consumption increases due to the additional amplifier, all design specifications are satisfied within the available voltage headroom and area constraints. The gain-boosted architecture therefore provides a robust solution for low-impedance, in-pixel current sensing.

```

--- Operating Point ---
V(vdd):      1.8          voltage
V(vtia):     1.31895     voltage
V(vm2):      1.15175     voltage
V(vcas):     0.701754    voltage
V(vb1):      0.2         voltage
V(n001):     1.30782     voltage
V(n002):     0.217765    voltage
V(vc):       0.7         voltage
V(vb2):      0.5         voltage
Id(M4):      1.30349e-006 device_current
Ig(M4):      0           device_current
Ib(M4):      -2.27763e-013 device_current
Is(M4):      -1.30349e-006 device_current
Id(M8):      6.45471e-007 device_current
Ig(M8):      -0          device_current
Ib(M8):      5.0218e-013  device_current
Is(M8):      -6.45472e-007 device_current
Id(M7):      6.58024e-007 device_current
Ig(M7):      -0          device_current
Ib(M7):      6.58252e-013 device_current
Is(M7):      -6.58024e-007 device_current
Id(M1):      2.16994e-008 device_current
Ig(M1):      0           device_current
Ib(M1):      -7.1189e-013 device_current
Is(M1):      -2.16987e-008 device_current
Id(M6):      6.45472e-007 device_current
Ig(M6):      0           device_current
Ib(M6):      -5.62978e-012 device_current
Is(M6):      -6.45467e-007 device_current
Id(M5):      6.58024e-007 device_current
Ig(M5):      0           device_current
Ib(M5):      -1.87116e-012 device_current
Is(M5):      -6.58022e-007 device_current
Id(M2):      9.18768e-008 device_current
Ig(M2):      0           device_current
Ib(M2):      -2.04074e-012 device_current
Is(M2):      -9.18748e-008 device_current
Id(M3):      -9.18768e-008 device_current
Ig(M3):      -0          device_current
Ib(M3):      4.91048e-013 device_current
Is(M3):      9.18763e-008 device_current
I(C1):       3.50877e-025 device_current
I(C1):       1.31895e-024 device_current
I(I1):       0           device_current
I(R1):       7.01754e-008 device_current
I(V4):       0           device_current
I(Vref):     0           device_current
I(V2):       0           device_current
I(Vdd):      -1.39537e-006 device_current

```

Figure 8: DC operating point voltages of the gain-boosted TIA.

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Semiconductor Device Operating Points:
--- BSIM3 MOSFETS ---

Name:      m4      m8      m7      m1      m6
Model:     nch.8   pch.5   pch.5   nch.2   nch.5
Id:        1.30e-06 6.45e-07 6.58e-07 2.17e-08 6.45e-07
Vgs:       5.00e-01 0.00e+00 1.56e-01 2.00e-01 4.82e-01
Vds:       2.18e-01 4.92e-01 6.48e-01 7.02e-01 1.09e+00
Vbs:       0.00e+00 4.92e-01 6.48e-01 0.00e+00 -2.18e-01
Vth:       4.91e-01 -4.61e-01 -4.61e-01 4.90e-01 5.50e-01
Vdsat:     7.91e-02 -8.33e-02 -8.34e-02 4.61e-02 5.57e-02
Gm:        2.40e-05 1.12e-05 1.14e-05 6.34e-07 1.43e-05
Gds:       3.86e-07 8.34e-08 7.81e-08 3.21e-09 7.98e-08
Gmb:       6.87e-06 3.77e-06 3.83e-06 1.91e-07 3.69e-06
Cbd:       0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.00e+00
Cbs:       0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.00e+00
Cgssov:    3.59e-16 6.46e-16 6.46e-16 7.32e-15 7.26e-16
Cgdov:     3.59e-16 6.46e-16 6.46e-16 7.32e-15 7.26e-16
Cgbov:     9.02e-17 1.36e-16 1.36e-16 9.02e-17 9.02e-17
dQgdVgb:   2.74e-15 6.12e-15 6.12e-15 3.43e-14 3.59e-15
dQgdVdb:   -3.61e-16 -6.46e-16 -6.46e-16 -7.32e-15 -7.26e-16
dQgdVsb:   -1.71e-15 -4.39e-15 -4.39e-15 -7.33e-15 -1.24e-15
dQddVgb:   -3.75e-16 -6.55e-16 -6.51e-16 -7.32e-15 -7.26e-16
dQddVdb:   3.65e-16 6.48e-16 6.47e-16 7.32e-15 7.26e-16
dQddVsb:   1.46e-17 9.89e-18 5.91e-18 1.23e-20 2.91e-19
dQbdVgb:   -7.16e-16 -1.41e-15 -1.41e-15 -1.96e-14 -1.64e-15
dQbdVdb:   -1.32e-18 -3.29e-19 -4.78e-20 2.29e-22 2.91e-20
dQbdVsb:   -3.40e-16 -8.13e-16 -8.12e-16 -1.28e-18 -1.17e-16

Name:      m5      m2      m3
Model:     nch.5   nch.5   pch.7
Id:        6.58e-07 9.19e-08 -9.19e-08
Vgs:       4.84e-01 4.50e-01 -4.81e-01
Vds:       9.34e-01 6.17e-01 -4.81e-01
Vbs:       -2.18e-01 -7.02e-01 0.00e+00
Vth:       5.50e-01 6.56e-01 -4.38e-01
Vdsat:     5.61e-02 4.75e-02 -8.89e-02
Gm:        1.46e-05 2.64e-06 1.54e-06
Gds:       8.40e-08 1.44e-08 3.54e-09
Gmb:       3.75e-06 5.72e-07 5.05e-07
Cbd:       0.00e+00 0.00e+00 0.00e+00
Cbs:       0.00e+00 0.00e+00 0.00e+00
Cgssov:    7.26e-16 3.66e-15 3.18e-16
Cgdov:     7.26e-16 3.66e-15 3.18e-16
Cgbov:     9.02e-17 9.02e-17 5.46e-16
dQgdVgb:   3.61e-15 1.44e-14 1.06e-14
dQgdVdb:   -7.26e-16 -3.66e-15 -3.19e-16
dQgdVsb:   -1.27e-15 -3.70e-15 -8.16e-15
dQddVgb:   -7.26e-16 -3.66e-15 -3.39e-16
dQddVdb:   7.26e-16 3.66e-15 3.23e-16
dQddVsb:   4.09e-19 6.69e-20 2.32e-17
dQbdVgb:   -1.63e-15 -7.01e-15 -2.56e-15
dQbdVdb:   2.84e-20 5.06e-22 -7.08e-19
dQbdVsb:   -1.22e-16 -4.03e-18 -1.98e-15

Date: Tue Jan 27 18:03:45 2026
Total elapsed time: 0.084 seconds.

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Figure 9: Detailed device operating point and small-signal parameters.

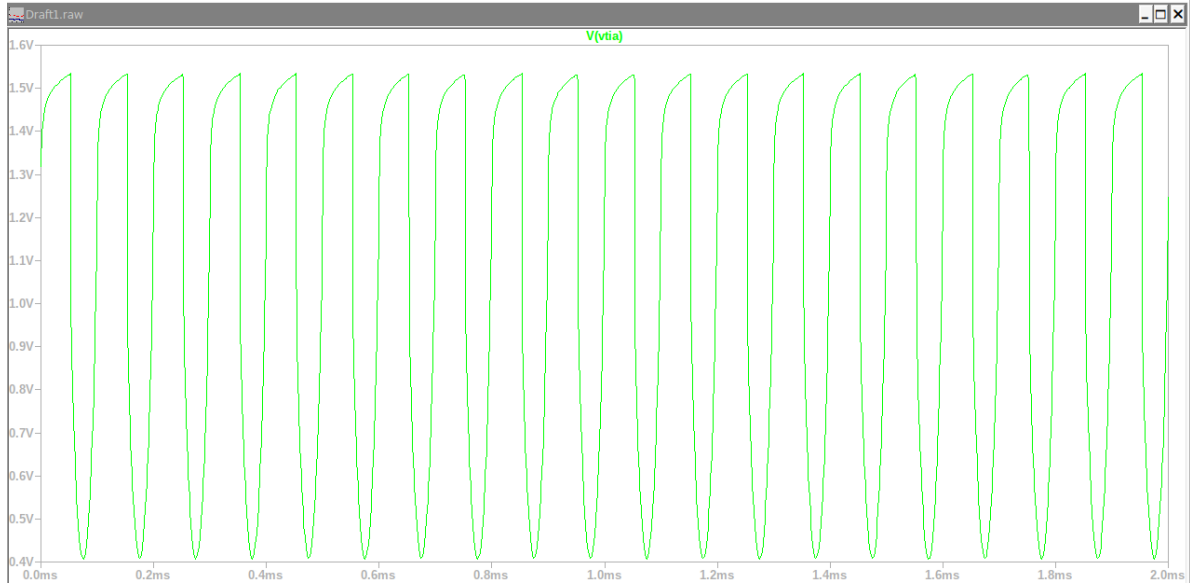


Figure 10: AC transimpedance gain magnitude of the gain-boosted TIA.

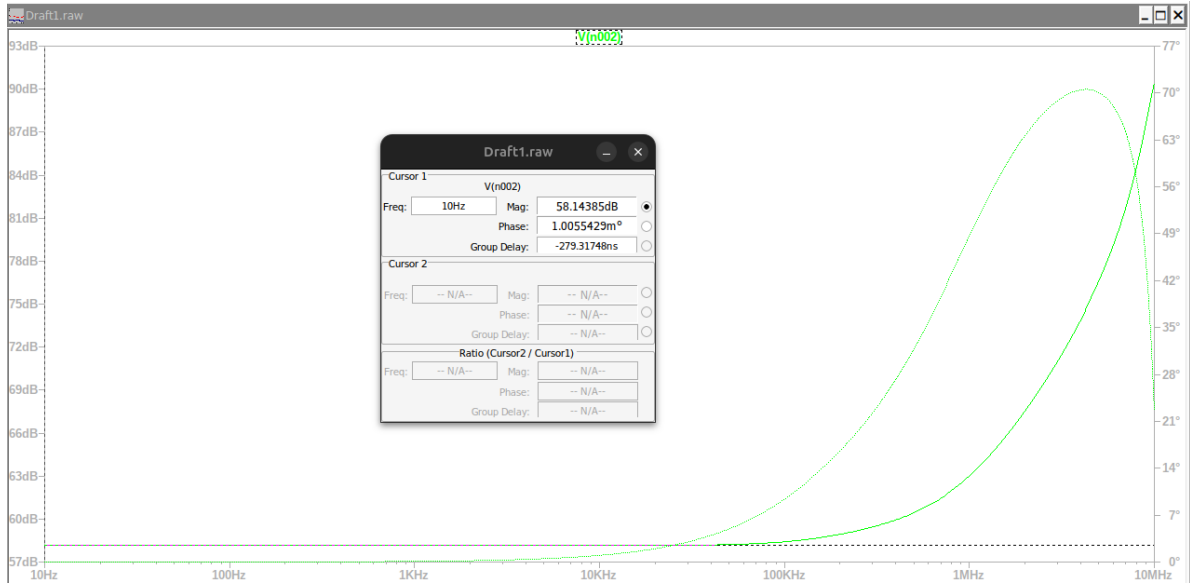


Figure 11: Input impedance magnitude of the gain-boosted TIA.

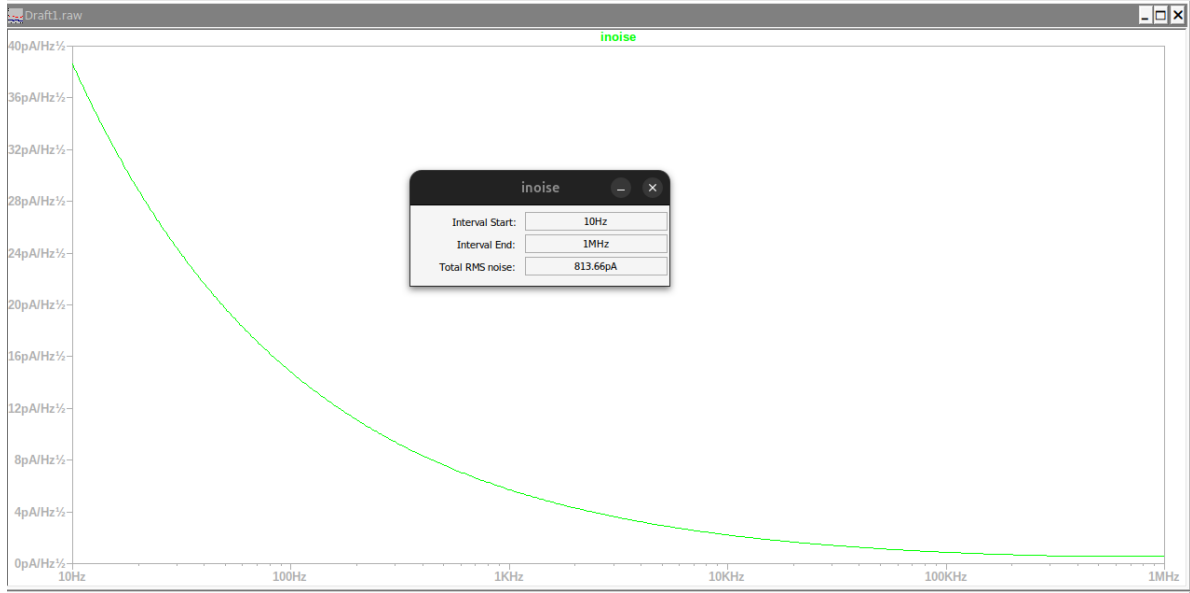


Figure 12: Input-referred current noise spectral density and integrated RMS noise.

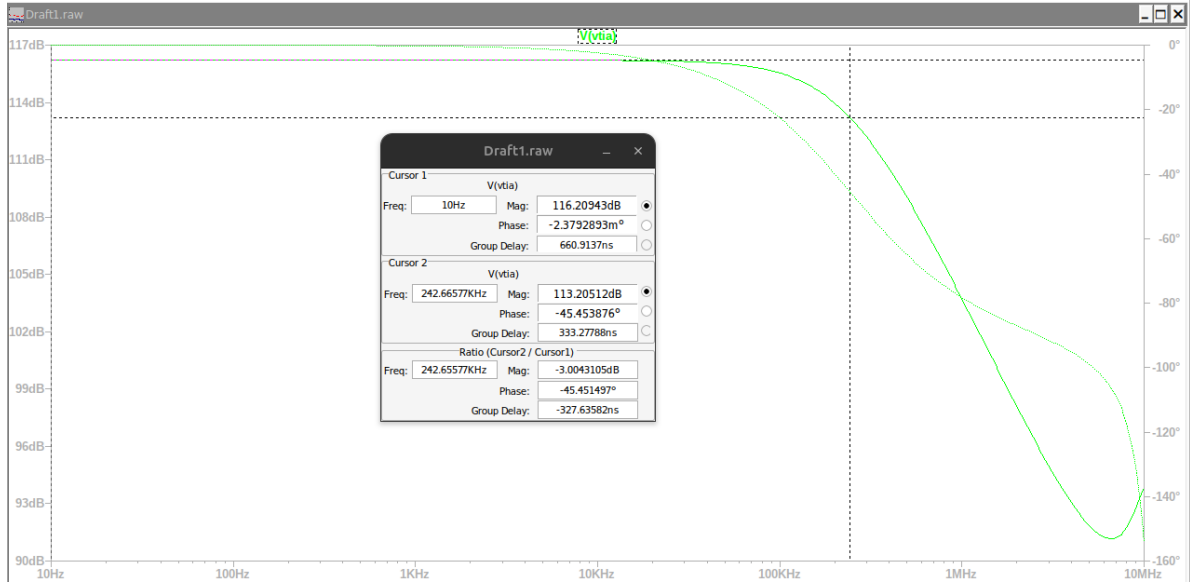


Figure 13: Transient response of V_{TIA} for $I_{in} = 10 \mu A$ at 10 kHz.